

ZLATIBOR, Serbia

WMO: 133670

Lat: **43.733N**Lon: **19.717E**Elev: **1029**StdP: **89.56**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.4	-11.0	-15.3	1.1	-13.4	-12.7	1.4	-10.5	7.5	6.4	6.5	4.8	1.6	230	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	29.1	18.1	27.2	17.8	25.6	17.3	19.6	25.9	18.8	25.0	18.0	24.1	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	14.2	23.0	16.5	13.4	22.0	15.7	12.6	21.0	60.9	26.0	57.8	25.0	55.0	24.1	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.1	4.2	DB	-16.1	31.6	2.6	2.1	-18.0	33.1	-19.5	34.3	-21.0	35.5	-22.9	37.0	(4)
(5)			WB	-16.5	21.4	2.6	0.9	-18.3	22.1	-19.8	22.6	-21.3	23.0	-23.2	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-1.5	-0.3	3.3	8.3	12.7	16.4	18.4	18.7	13.9	9.4	4.9	-0.6	(6)	
	DBStd	8.53	5.31	5.59	5.09	4.53	4.08	4.02	3.57	3.65	4.26	4.95	5.32	4.94	(7)	
	HDD10.0	1544	356	288	213	81	21	3	0	1	13	73	164	330	(8)	
	HDD18.3	3657	614	521	467	301	176	82	44	39	144	276	403	588	(9)	
	CDD10.0	1064	0	1	4	30	106	196	259	271	129	56	12	1	(10)	
	CDD18.3	134	0	0	0	0	3	24	46	51	9	1	0	0	(11)	
	CDH23.3	969	0	0	0	1	24	173	341	369	60	2	0	0	(12)	
	CDH26.7	212	0	0	0	0	1	24	77	99	11	0	0	0	(13)	
(14) Wind	WSAvg	1.9	2.1	2.1	2.1	2.1	1.9	1.7	1.7	1.6	1.7	1.7	2.1	2.2	(14)	
(15) Precipitation	PrecAvg	951	65	58	61	75	104	109	101	70	83	68	83	74	(15)	
	PrecMax	1157	132	124	110	135	155	238	168	131	222	175	143	146	(16)	
	PrecMin	731	15	18	15	10	58	42	41	19	17	10	25	11	(17)	
	PrecStd	119	37	31	27	27	31	45	33	35	55	44	32	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.3	17.8	22.5	25.9	29.0	31.4	31.9	28.3	22.5	18.3	12.5	(19)	
		MCWB	6.5	7.9	9.4	12.5	15.4	19.0	17.9	18.2	16.5	14.5	11.4	8.6	(20)	
	2%	DB	8.9	11.1	14.9	19.5	23.4	27.0	28.8	29.4	24.9	20.4	15.5	10.0	(21)	
		MCWB	5.5	6.1	8.0	11.1	15.1	18.4	18.3	17.9	15.7	13.5	10.1	6.7	(22)	
	5%	DB	7.3	9.2	12.8	17.1	21.5	25.4	26.9	27.2	22.4	18.3	13.6	8.2	(23)	
		MCWB	4.4	5.1	6.9	9.9	14.1	17.8	18.1	17.4	15.3	12.7	9.0	5.5	(24)	
	10%	DB	5.6	7.2	10.7	15.1	19.6	23.6	25.1	25.2	20.3	16.3	11.9	6.2	(25)	
		MCWB	3.3	4.1	6.0	9.0	13.3	17.0	17.5	17.1	14.3	11.8	8.3	4.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	8.2	9.9	13.3	17.1	20.7	20.7	20.5	17.8	15.4	12.2	8.9	(27)	
		MCDB	10.3	13.1	16.5	20.5	23.4	26.7	26.9	27.2	24.8	21.1	17.1	11.7	(28)	
	2%	WB	5.7	6.5	8.6	11.7	15.7	19.3	19.6	19.3	16.6	14.2	10.7	7.0	(29)	
		MCDB	8.5	10.6	13.8	18.4	21.6	25.3	25.9	25.7	23.0	18.8	14.5	9.6	(30)	
	5%	WB	4.7	5.4	7.5	10.5	14.7	18.3	18.7	18.4	15.7	13.1	9.5	5.7	(31)	
		MCDB	7.0	8.7	12.1	15.9	20.4	24.0	25.0	24.9	21.5	17.4	13.0	8.1	(32)	
	10%	WB	3.5	4.2	6.3	9.5	13.7	17.3	17.9	17.6	14.8	12.2	8.4	4.3	(33)	
		MCDB	5.4	7.0	10.4	14.0	18.9	22.9	24.0	23.8	19.6	15.8	11.4	6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.9	7.7	8.2	8.8	9.1	9.4	9.4	8.2	7.9	6.8	5.6	(35)	
		MCDBR	7.5	9.4	11.1	11.3	11.6	11.6	11.9	12.1	11.2	10.3	9.0	7.7	(36)	
	5% WB	MCWBR	4.8	5.6	6.1	5.7	5.7	5.8	5.4	5.0	5.3	5.4	5.3	5.3	(37)	
		MCWBR	7.0	8.8	10.1	9.9	10.5	10.7	10.7	10.3	10.0	9.4	8.1	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.315	0.351	0.399	0.396	0.388	0.390	0.396	0.376	0.341	0.308	0.289	(40)	
		taud	2.431	2.345	2.312	2.211	2.263	2.318	2.308	2.290	2.337	2.439	2.478	2.455	(41)	
	Ebn at Noon	Ebn at Noon	843	876	884	864	876	882	875	857	847	836	815	809	(42)	
		Edh at Noon	74	95	113	135	132	126	126	124	109	87	70	66	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.44	2.22	3.37	4.26	5.21	5.90	6.10	5.53	3.85	2.72	1.71	1.22	(44)	
	RadStd	RadStd	0.20	0.25	0.40	0.53	0.49	0.57	0.48	0.51	0.42	0.33	0.19	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.80	N/A	N/A	+1.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	+0.62	N/A	N/A	N/A	+0.36	N/A	N/A	-119	+100	N/A	N/A

Nomenclature: See separate page